

PFISTERER

Instructions for Use

Earthing sticks



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1 General

These instructions for use enable safe and efficient use of the device. They are a constituent part of the device and must be kept in the immediate vicinity of the device such that they are accessible to personnel at all times. The personnel must have carefully read and understood these instructions before starting any work. Strict compliance with the safety instructions and working procedures provided in these instructions is a fundamental requirement for work safety.

In addition, the local accident prevention regulations and general safety requirements for the scope of application of the device also apply.

1.1 Limitation of liability

All information and data contained in these instructions for use were compiled under consideration of the applicable standards and regulations, technological state-of-the-art and long standing expertise and experience.

The manufacturer shall accept no liability whatsoever for damage attributed to:

- Non-compliance with these instructions
- Use not for intended purpose
- Use of untrained/unskilled personnel
- Unauthorised modifications
- Technical modifications
- Use of non-approved spare parts

Due to fact that special versions or additional options may have been ordered or innovative technical modifications may have been made, the actual scope of delivery can differ from the explanations and illustrations described in these instructions.

The obligations specified in the delivery agreement, general terms and conditions of business as well as the manufacturer's terms and conditions of delivery and the legal regulations applicable at the time the contractual agreement was concluded shall apply.

1.2 Important information

**WARNING!****Risk of injury posed by insufficiently qualified personnel!**

Incorrect use can cause serious personal injury and considerable material damage. Therefore:

- All activities must only be carried out by qualified personnel.

Earthing sticks are used to introduce earthing and short-circuiting facilities to parts of electrical power installations for earthing and short-circuiting purposes. The earthing stick consists of the insulating part, black ring, handle and the coupling for accepting a connector.

The operating and maintenance instructions provided below must be complied with. The earthing sticks conform to DIN EN 61235. The earthing stick may:

- only be used by technical or instructed personnel as defined by DIN VDE 0105 Part 1 for earthing and short-circuiting purposes.
- only be subjected to the maximum permissible head load (kg) as specified on the earthing stick.
- only be held by the handle. The handle is separated from the insulating part by a black ring. Do not grasp the stick beyond the black ring.
- only be used indoors and outdoors only when there is no precipitation or mist/fog.

The earthing stick must be checked by the user for obvious damage before use. Defective earthing sticks must not be used.

**WARNING!****Unauthorised persons are at risk!**

Unauthorised persons who do not satisfy the requirements specified in these instructions are not aware of the dangers in the work area. Therefore:

- Keep unauthorised persons away from the work area.
- Stop work as long as there are unauthorised persons within the work area.

2 Safety

2.1 Responsibility of the operating company

The device is used in industrial applications. The operating company is therefore subject to the legal obligations relating to occupational health and safety.

In addition to the health and safety information contained in these instructions, the safety, accident prevention and environmental regulations applicable for the scope of application of the device must be observed. The following applies in particular:

- The operating company must be familiar with the applicable occupational health and safety regulations and determine additional hazards, resulting from the specific local operating conditions, by compiling a risk analysis. These requirements must be transcribed into instructions for operating the device.
- The operating company must continually check throughout the entire service life of the device whether the self-compiled operating instructions still conform to current regulations and amend them if necessary.
- The operating company must unambiguously govern and delegate responsibilities for installation, operation, maintenance and cleaning.
- The operating company must ensure that all employees who work with the device have read and understood these instructions. The operating company must also provide training for the employees at regular intervals and inform them of the risks involved.
- The operating company must provide personnel with the requisite safety equipment.

Furthermore, the operating company is responsible for ensuring that the device is always in technically perfect working order. The following therefore applies:

- The operating company must ensure that the maintenance intervals specified in these instructions are strictly adhered to.
- The operating company must have regular inspections carried out to ensure that all safety devices and equipment used in the system in which the device is installed are in place and fully functional.

3 Technical data

3.1 Scope of application

Earthing sticks

Description	Value	Unit
Overall length		
one-piece	1000 – 3500	mm
telescopic	1500 – 5900	mm
Max. head load	35 – 10	kg

3.2 Operating conditions

Ambient conditions

Description	Value	Unit
Temperature range	-25 to +55	°C
Max. relative humidity	60	%

3.3 Storage

Storing packaged units

Store packed units under following conditions:

- Do not keep outdoors
- Store in a dry and dust-free place
- Do not expose to aggressive media
- Protect from direct sunlight
- Storage temperature: -25 to +55 °C
- Relative humidity: max. 60 %
- If stored for longer than 3 months regularly check the general condition of all components and the packaging.

4 Design and functional principle

4.1 Overview

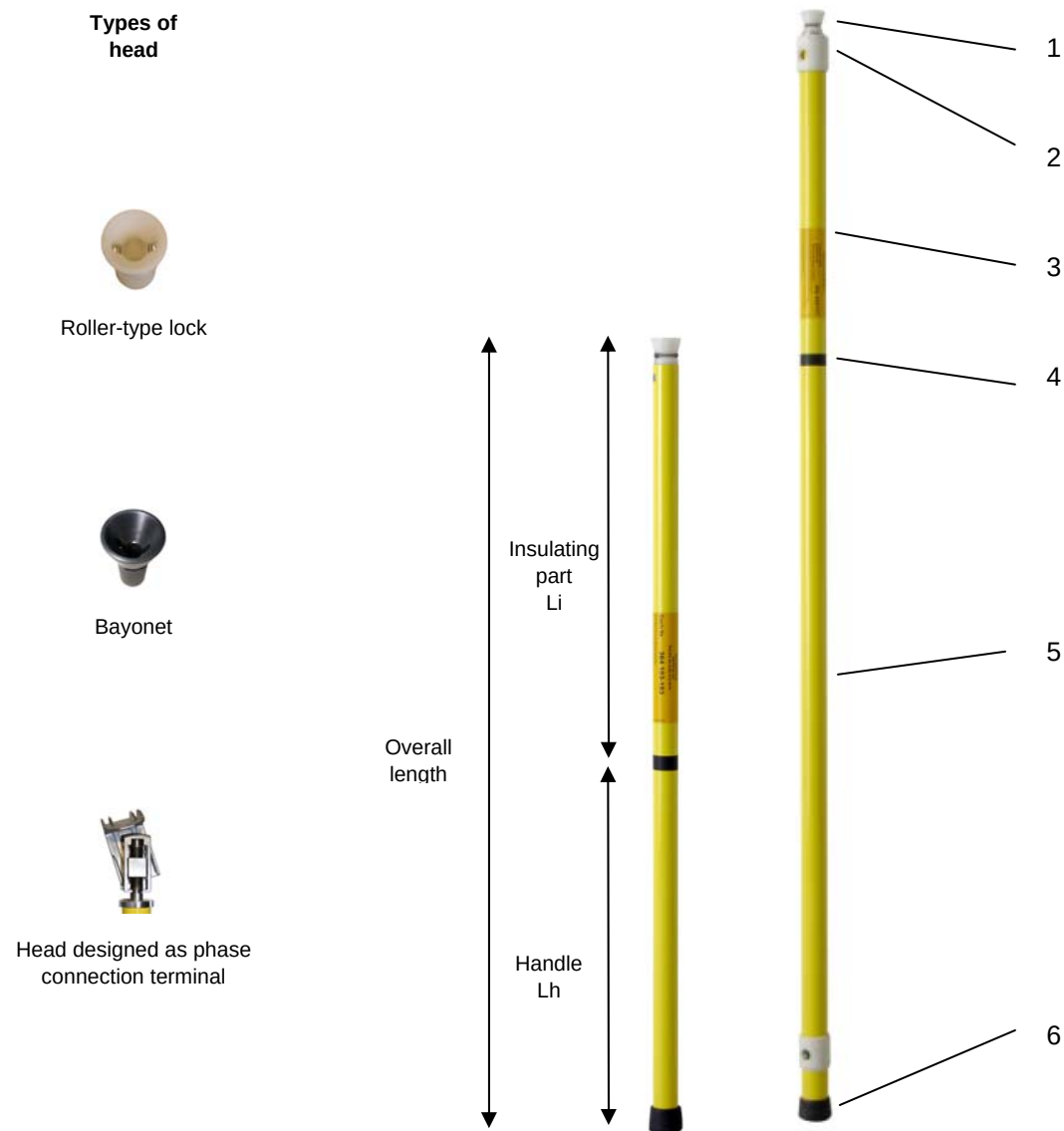


Fig. 1: Earthing stick

- | | | | |
|---|--------------------------------|---|----------------------|
| 1 | Working head | 4 | Black ring |
| 2 | Lock (only telescopic version) | 5 | Handle, length L_h |
| 3 | Insulating part, length L_i | 6 | End piece |

Earthing sticks are available with different heads corresponding to requirements. Their method of use may therefore vary slightly from that described in these instructions.

5 Using the earthing stick



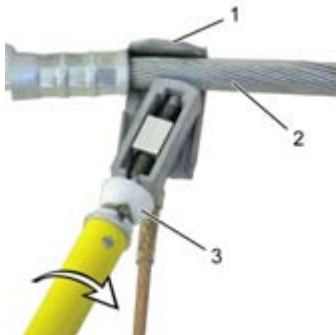
1. Fitting the phase connection terminal

1.1. Roller-type lock

Insert phase connection terminal with stem in the working head of the earthing stick and press down until the cross pins snap into the roller-type lock on the earthing stick.

1.2. Bayonet head

Insert phase connection terminal with stem in the working head of the earthing stick and turn clockwise by 90° until the stem locks in.



2. With the aid of the earthing stick (3), attach the phase connection terminal (1) to the conductor cable (2) and tighten by turning the earthing stick in clockwise direction. Check that the phase connection terminal is securely seated.

3. Remove earthing stick.

4. Repeat steps 1 to 3 to connect further phase connection terminals.

6 Maintenance and care

The operating company must have the necessary maintenance work carried out immediately if increased wear is determined on the individual components or assemblies is determined during routine inspections.

Interval	Maintenance work	Performed by
Before use	Visually inspect the condition of the device and check for damage	Qualified electrician
After use	Clean device. If soiled, wipe the earthing rod with a soft, moist cloth. Note: – Do not use aggressive cleaning agents. – After cleaning, make sure that all previously opened covers are closed correctly again. – Make sure that the device is completely dry before it is used again.	Qualified electrician